Traffic micromosimulation

Broadly speaking, there are two major classes of traffic microsimulation models:

1. Cellular automata. Discrete in space and time.
2. Car-following models. Continuous in space and time.

Highway traffic



Nagel-Schreckenberg Model

A lane on a highway can be simulated as a 1-dimensional cellular automaton where cars jump between discrete cells at discrete time steps.

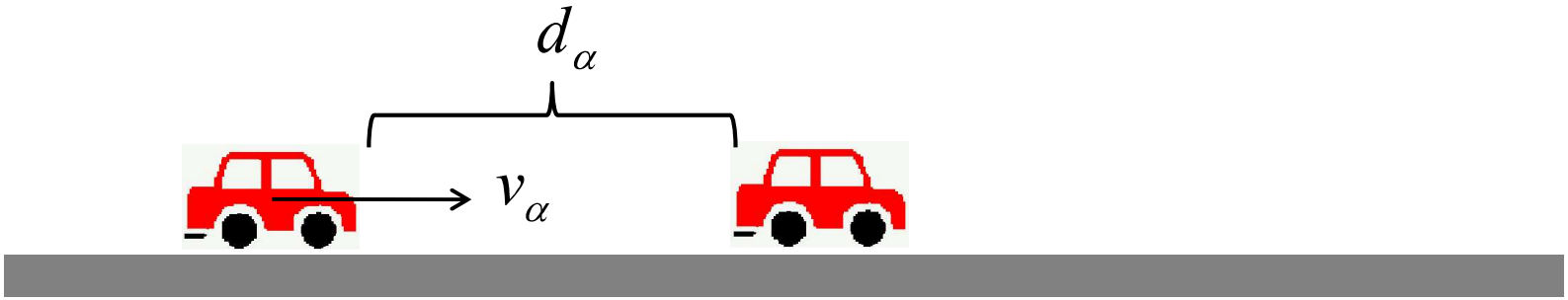
1) **Acceleration:** if the velocity v of a vehicle is lower than $v_{\max}$ and if the distance to the next car ahead is larger than $v+1$, the speed is advanced by one $[v \to v+1]$.
2) **Slowing down (due to other cars):** if a vehicle at site i sees the next vehicle at site $i+j$ (with $j \leq v$), it reduces its speed to $j-1$ $[v \to j-1]$.
3) **Randomization:** with probability p, the velocity of each vehicle (if greater than zero) is decreased by one $[v \to v-1]$.
4) **Car motion:** each vehicle is advanced v sites.



Car-following models

The Optimal Velocity Model (Bando et al, 1995) is one of the simplest models within the class of *car-following models*:

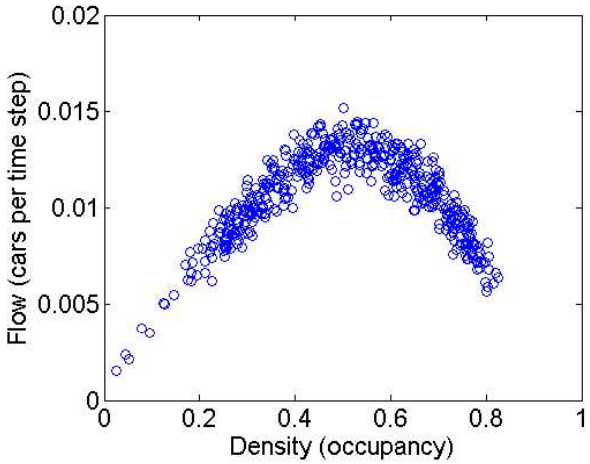
$$\frac{dv_\alpha}{dt} = \frac{v'_e(d_\alpha) - v_\alpha}{\tau} \qquad v'_e(d_\alpha) = \frac{v_0}{2}\left[\tanh(d_\alpha - d_c) + \tanh(d_c)\right]$$



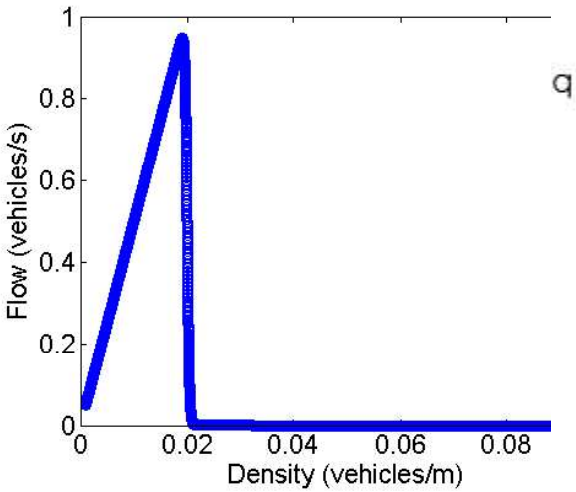
State of car: Position *x(t)*, Velocity $v_\alpha(t)$, desired velocity v_0

Model comparison

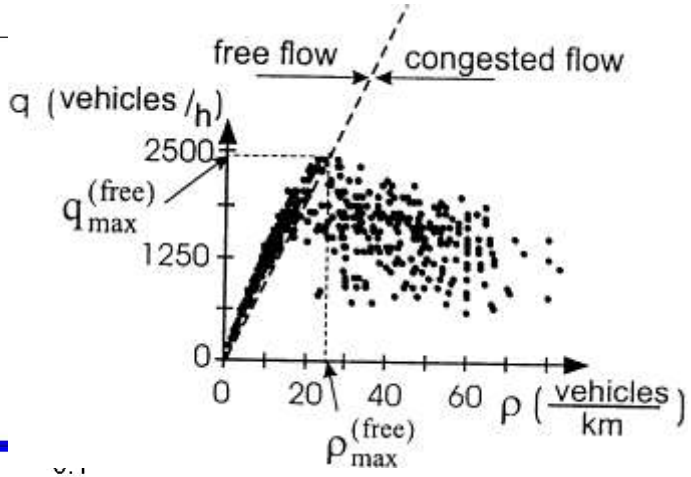
Comparison of fundamental diagrams: Simulated and empirical:



CA traffic model



$$v'_e(d_\alpha) = \frac{v_0}{2}\left[\tanh(d_\alpha - d_c) + \tanh(d_c)\right]$$



Empirical Fundamental Diagram:
B S Kerner (2000) Traffic and Granular Flow '99, pp 253-283